

PERSONAL EXPENSE TRACKER

Project Summary

The **Personal Expense Tracker** is a GUI-based financial management application developed using **Python, Tkinter, and SQLite**. The main purpose of the system is to help users efficiently record, manage, and analyze their personal income and expenses.

The application allows users to **add, edit, delete, and view income and expense transactions** along with details such as date, category/source, amount, payment method, and description. All transaction records are stored in an **SQLite database**, providing organized and persistent data management.

The system also provides a **dashboard** that displays important financial information such as total income, total expenses, current balance, monthly expenses, savings rate, monthly financial summaries, and recent transactions. For visual analysis, **Matplotlib** is used to generate charts such as expense distribution by category, income-versus-expense comparison, and expense trends.

A **reporting module** allows users to filter transactions according to date, type, category, and search terms. Reports can also be exported in **CSV and Excel formats using OpenPyXL**, making the financial data easier to analyze and maintain.

The project further focuses on upgrading basic expense tracking into a more **intelligent financial analysis system**. Proposed enhancements include **unusual spending detection, expense forecasting, smart budget recommendations, month-to-month comparison, and actionable financial insights**. These features aim to transform stored transaction data into useful information that can help users understand their spending patterns and improve financial planning.

Overall, the project combines **transaction management, database storage, visualization, reporting, and intelligent analytics** into a single desktop application, providing a structured approach to personal financial management.